Jishnu Nambissan Thekkeppattu (he/him)

Contact Information

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Research Interests

Radio astronomy, radio interferometry, instrumentation and techniques. Phased arrays, calibration and array signal processing, wide-field polarimetry. High speed DSP, reconfigurable computing, FPGAs. Experimental and observational 21-cm cosmology, precision radiometry, data analysis for 21-cm experiments, radio-sky modelling at low-frequencies. All-sky spherical harmonic transit interferometry (m-mode imaging). Numerical linear algebra, ill-posed inverse problems, iterative methods. Ionospheric scintillation.

Professional experience

Aug 2025 - Present	CERC Fellow, CSIRO Space and Astronomy, Australia
Aug 2022 - Jul 2025	Research Associate, Curtin University, Australia
Nov 2016 - Jul 2022	Research Scientist/Research Associate, Raman Research Institute, India
Jan 2016 - Nov 2016	Assistant System Engineer, Tata Consultancy Services, India

Education

Curtin University, Australia

Doctor of Philosophy – Physics (2019–2022)

- Thesis title : Towards detection of redshifted 21-cm signal from Cosmic Dawn and Epoch of Reionization.
- Thesis supervisors : Dr. Cathryn M. Trott, Dr. Benjamin McKinley (Curtin Institute of Radio Astronomy, Bentley, WA)

Kannur University, India

Bachelor of Technology in Electronics and Communication Engineering (2011–2015)

- Aggregate percentage : 81.2%
- Undergraduate project : A VHF board for Software Defined Radio (SDR) applications.
- Project advisor: Dr. V. Vinodkumar (Govt. College of Engineering, Kannur)

Publications

ORCID : 0000-0002-3079-4046

First authored, peer reviewed, journal papers

Jishnu N. Thekkeppattu, Randall Wayth, and Marcin Sokolowski, *Calibration of an SKA-Low prototype station using holographic techniques*, 2024, Radio Science https://doi.org/10.1029/2023RS007847.

Jishnu N. Thekkeppattu, Cathryn M. Trott, and Benjamin McKinley, *Singular spectrum analysis of time series data from low frequency radiometers, with an application to SITARA data*, 2023, Monthly Notices of the Royal Astronomical Society, https://doi.org/10.1093/mnras/stad522.

Jishnu N. Thekkeppattu, Benjamin McKinley, Cathryn M. Trott, Jake Jones, Daniel C. X. Ung, *System design and calibration of SITARA - a global 21-cm short spacing interferometer prototype*, 2022, Publications of the Astronomical Society of Australia, https://doi.org/10.1017/pasa.2022.13

Jishnu Nambissan, T.; Ravi Subrahmanyan; R. Somashekar; N. Udaya Shankar; Saurabh Singh; A. Raghunathan; B. S. Girish; K. S. Srivani; Mayuri Sathyanarayana Rao *SARAS 3 CD/EoR radiometer: design and performance of the receiver*, 2021, Experimental Astronomy, https://doi.org/10.1007/s10686-020-09697-2

Co-authored, peer reviewed, journal papers

Macario, Giulia; Pupillo, Giuseppe; Bernardi, Gianni; Di Ninni, Paola; Comoretto, Giovanni; Bolli, Pietro; Mattana, Andrea; Monari, Jader; Perini, Federico; Schiaffino, Marco; Sokolowski, Marcin; Wayth, Randall; **Thekkeppattu, Jishnu N.**, *Intrinsic cross-polarization ratio maps from all-sky observations with the SKA-Low prototype station Aperture Array Verification System 2*, Journal of Astronomical Telescopes, Instruments, and Systems, Volume 11, id. 028008 (2025)
https://doi.org/10.1117/1.JATIS.11.2.028008

Saurabh Singh, **Jishnu Nambissan T.**, Ravi Subrahmanyan, N. Udaya Shankar, B. S. Girish, A. Raghunathan, R. Somashekar, K. S. Srivani, Mayuri Sathyanarayana Rao, *On the detection of a cosmic dawn signal in the radio background*, Nature Astronomy, 6, 607-617 (2022) https://doi.org/10.1038/s41550-022-01610-5

Girish, B. S.; Srivani, K. S.; Subrahmanyan, Ravi; Udaya Shankar, N.; Singh, Saurabh; **Jishnu Nambissan, T.**; Sathyanarayana Rao, Mayuri; Somashekar, R.; Raghunathan, A. *SARAS CD/EoR Radiometer: Design and Performance of the Digital Correlation Spectrometer*, 2020, Journal of Astronomical Instrumentation, Volume 9, Issue 2. https://doi.org/10.1142/S2251171720500063

B McKinley, C M Trott, M Sokolowski, R B Wayth, A Sutinjo, N Patra, **J. Nambissan T.**, D C X Ung, The All-Sky SignAl Short-Spacing INterferometer (ASSASSIN) - I. Global-sky measurements with the Engineering Development Array-2, Monthly Notices of the Royal Astronomical Society, Volume 499, Issue 1, November 2020, Pages 52-67, https://doi.org/10.1093/mnras/staa2804

Raghunathan, A.; Subrahmanyan, R; Shankar, N. U.; Singh, S; **Nambissan, J.**; Kavitha, K.; Mahesh, N.; Somashekar, R.; Sindhu, G; Girish, B. S.; Srivani, K. S.; Rao, M. S., *A Floating Octave Bandwidth Cone-Disk Antenna for Detection of Cosmic Dawn*, in IEEE Transactions on Antennas and Propagation, vol. 69, no. 10, pp. 6209-6217, Oct. 2021, https://doi.org/10.1109/TAP.2021.3069563

Adrian T. Sutinjo, Benjamin McKinley, Leonid Belostotski, Daniel C. X. Ung, and **Jishnu Nambissan T.**, *Design Equations for Closely Spaced Two-Element Interferometer for Radio Cosmology*, URSI Radio Science Letters, Vol. 2, 2020,
https://doi.org/10.46620/20-0017

Conference Proceedings

Thekkeppattu, J. N. (2026). *Monitoring equatorial ionospheric scintillation using low-cost software defined radios - some results from the Indian sector.* In Proceedings of the XXXVIth URSI General Assembly and Scientific Symposium - URSI GASS 2026. XXXVIth URSI General Assembly and Scientific Symposium. URSI - International Union of Radio Science.
https://doi.org/10.46620/URSIGASS26/IVTK6571

Thekkeppattu, J. N., Humphrey, D., Thyagarajan, N. (2026). *Ultrafast real-time imaging with large-N aperture arrays using FPGA accelerators.* In Proceedings of

the XXXVIth URSI General Assembly and Scientific Symposium - URSI GASS 2026. XXXVIth URSI General Assembly and Scientific Symposium. URSI - International Union of Radio Science.
https://doi.org/10.46620/URSIGASS26/CUUS2167

Thekkeppattu, J. N., Cook, J. H., Wayth, R. B., Carozzi, T. D., Sokolowski, M., Kriele, M. A., and Vijayaraghavan, M. (2025). *All-sky m-mode map making at 159 MHz with EDA2 and SE607HBA LOFAR station.* In Proceedings of the 7th URSI Asia-Pacific Radio Science Conference - AP-RASC 2025. 7th URSI Asia-Pacific RadioScience Conference. URSI - International Union of Radio Science.
https://doi.org/10.46620/ursiaprasc25/wqun6574

Cook, J., **Thekkeppattu, J.**, Wayth, R., and Kriele, M. (2025). *All Sky maps from m-mode Spherical Harmonic Transit Interferometry with the MWA and the EDA2.* In Proceedings of the 7th URSI Asia-Pacific Radio Science Conference - AP-RASC 2025. 7th URSI Asia-Pacific Radio Science Conference. URSI - International Union of Radio Science.
https://doi.org/10.46620/ursiaprasc25/pomb8661

Thekkeppattu, J. N., Sokolowski, M., and Wayth, R. B. (2025). *Broadband self-holography of EDA2 - an SKA-Low prototype station.* In Proceedings of the 7th URSI Asia-Pacific RadioScience Conference - AP-RASC 2025. 7th URSI Asia-Pacific Radio Science Conference. URSI - International Union of Radio Science.
https://doi.org/10.46620/ursiaprasc25/hech4944

Raghunathan, Agaram; Girish, B.S.; Somashekar, R.; Srivani, K.S.; Singh, Saurabh; Subrahmanyan, Ravi; Shankar, N. Udaya; Rao, Mayuri Sathyanarayana; **Nambissan T., Jishnu**, *Wide band antenna with Ultra-smooth spectral characteristics*, 2020 International Applied Computational Electromagnetics Society Symposium (ACES), 2020, pp. 1-2, https://doi.org/10.23919/ACES49320.2020.9196075

Nambissan, T. J., Nikhil, T., Vinodkumar, V., *A VHF Radio for Software Defined Radio Applications*, 2016, Procedia Technology, 24, 820, International Conference on Emerging Trends in Engineering, Science and Technology (ICETEST - 2015) https://doi.org/10.1016/j.protcy.2016.05.109

Non peer reviewed articles / pre-prints

Jishnu N. Thekkeppattu, *First results from a low cost software defined radio system monitoring VHF equatorial ionospheric scintillations in the Indian sector*, 2024, ESS open archive.

Research Experience

2025-Present	Development of an ultrafast radio camera using FPGAs Supervisor(s): Dr. Nithyanandan Thyagarajan, Dr. David Humphrey CSIRO Space and Astronomy, Marsfield, NSW.
2022-2025	SKA-Low bridging and Early Production Array activities & All-sky mapping with spherical harmonic transit interferometry Supervisor(s): Dr. Randall Wayth, Dr. David Davidson Curtin Institute of Radio Astronomy, Bentley, WA.
2019–2022	Towards detection of redshifted 21-cm signal from Cosmic Dawn and Epoch of Reionization. Supervisor(s): Dr. Cathryn Trott, Dr. Benjamin McKinley, Curtin Institute of Radio Astronomy, Bentley, WA.
2019–2021	Principal investigator of PRATUSH: Probing ReionizATion of the Universe using Signal from Hydrogen. (Funded for pre-project studies by the ISRO) Co-PIs : Dr. Saurabh Singh, Dr. Mayuri S. Rao Raman Research Institute, Bangalore, India.
2016–2021	On the detection of global 21-cm signal from Cosmic Dawn and Epoch of Reionization. Advisors: Dr. Ravi Subrahmanyan, Dr. N. Udaya Shankar, Raman Research Institute, Bangalore, India.
2019 June–August	Comparing LOFAR Epoch of Reionization target fields using 21-cm power spectra. Advisor : Dr. André Offringa, ASTRON, Dwingeloo, The Netherlands.
2014–2015	Reconfigurable radios using Field Programmable Gate Arrays. Advisor: Dr. V. Vinodkumar, Govt. College of Engineering, Kannur, India
2014–2015	A VHF board for Software Defined Radio (SDR) applications. Advisor: Dr. V. Vinodkumar, Govt. College of Engineering, Kannur, India
2013–2014	Multiplierless filter realisation using Canonic Signed Digits. Advisor: Dr. Manju Manuel, Govt. College of Engineering, Kannur, India

Selected talks

2026 Aug	*Monitoring equatorial ionospheric scintillation using low-cost software defined radios - some results from the Indian sector* - URSI-GASS-2026, held at Krakow, Poland (Young Scientist Award talk)
2026 Aug	*Ultrafast real-time imaging with large-N aperture arrays using FPGA accelerators* - URSI-GASS-2026, held at Krakow, Poland
2025 Dec	*Ultra-fast real-time imaging with large aperture arrays* - SALF-XI, held at Orleans, France (remote talk)
2025 Aug	*All-sky m-mode map making at 159 MHz with EDA2 and SE607HBA LOFAR station* - URSI-AP-RASC 2025, held at Sydney, Australia
2025 Aug	*Broadband self-holography of EDA2 - an SKA-Low prototype station* - URSI-AP-RASC 2025, held at Sydney, Australia
2025 Jun	*All-sky map making with spherical harmonics in the SKA-Low era* - SKAO General Science Meeting, A new era in astrophysics: Preparing for early science with the SKAO, held at Goerlitz, Germany
2020 Jan	*Updates from the SARAS experiment* - International School And Conference On Observing The First Billion Years Of The Universe Using The Next Generation Telescopes, held at IIT Indore, India (remote talk)

Honors and Awards

2026	Young Scientist Award, URSI-GASS-2026
2022	Joint winner of the ICRAR Ken and Julie Michael Prize
2014	Best student paper award in embedded systems and VLSI category at National Conference on Systems Energy and Environment - 14 held at Govt College of Engineering, Kannur
2010	Indian Space Research Organisation (ISRO) sponsored student delegate, 97th Indian Science Congress held at Kerala University.

Teaching

2025	COMT4003 - Mobile Radio Communications; Unit Co-ordinator & Lecturer (Curtin University)
2024	COMT6005 - Satellite Communications; Lecturer (Curtin University)
2024	COMT4003 - Mobile Radio Communications; Lecturer (Curtin University)
2023	COMT6005 - Satellite Communications; Lecturer (Curtin University)
2020	ELEN3002/6017 - Engineering Electromagnetic Labs (Curtin University)

Memberships

2023-Present	Junior member - International Astronomical Union (IAU)
2024-2025	Member - Institute of Electrical and Electronics Engineers (IEEE)
2013-2014	Student member - IEEE

Software tools and packages

Programming Languages:	python, C, C++, `bash` scripting
HDL:	VHDL
Software packages:	AMD Vivado, MATLAB, `miriad`
Operating systems:	Linux, Mac OS
Misc:	x86 and 8051 assembly, embedded C, arduino, Visual C# Verilog, EM simulators (NEC2, CST) and HPC systems

Observing and Field Experience

Deployment, operation and maintenance of a prototype two-element broadband interferometer SITARA at Inyarrimanha Ilgari Bundara - the CSIRO Murchison Radio-astronomy Observatory (2021-2022).

Measurement of global sky brightness temperature at low frequencies deploying SARAS at remote sites at Timbaktu Collective, India (2017-18), Indian Astronomical Observatory, Ladakh, India (2017-18) and Mavathur lake, India (2019).

Electromagnetic tests and measurements of performance and characteristics of the SARAS antennas and receivers at the Gauribidanur Field Station, India (2017-19) and Mavathur lake, India (2019).

Hardware and laboratory experience

Expertise in electronics and RF/microwave systems; using vector network analysers (Keysight FieldFox, PNA-X etc.) and their calibration, spectrum analysers, system level RF design, development and testing. Hands-on experience in handling fibre-optics, optical power measurements and RFoF systems. Experienced in performing measurements in an electromagnetic compatibility (EMC) chamber. Expertise in basic laboratory/workshop tasks such as soldering and miscellaneous mechanical works. FPGA systems (Zynq, Alveo v80).

Outreach

2025–Present	CSIRO S&A Marsfield colloquia organiser.
2020–2025	DIY radio telescope programme. In collaboration with the Curtin ASTRO-3D node, built a low-cost hobby radio telescope operating at 1420 MHz for outreach purposes.
2015	Institute of Electrical and Electronics Engineers - Teachers In Service programme (IEEE-TISP) volunteer. Conducted science and engineering outreach programmes for high school students.

Volunteering and Hobbies

2021-Present	Australian Red Cross Lifeblood blood, platelets & plasma donor
2014-Present	Amateur radio operator, holding Australian callsign VK6JN and Indian callsign VU3VWB